

 MSC DISSERTATION · KENYA COFFEE

Iteration 2

model log

Six methodological changes to the NARDL/XGBoost pipeline, the end-to-end run, and the iteration 2.1 break-dummy diagnostic.

PREPARED BY

Calvin Baraka Rimba

DATE

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STATUS

Run complete · iteration 3 deferred

Iteration 2 model log

MSc Dissertation — Kenya coffee NARDL/XGBoost pipeline. Six methodological changes, the end-to-end run, and the iteration 2.1 break-dummy diagnostic.

RMSE LOG GAP

0.017

–52% vs iteration 1 (0.036)

STAGE 1 BOUNDS F

18.07

▲ up from 16.95

STAGE 1 BG(12) P

0.0003

▼ still rejecting at 1%

NARDL_PRED_T SHAP

51%

of feature mass; dominates

SUMMARY

Iteration 2 implemented six methodological changes to the NARDL/XGBoost pipeline and ran end-to-end in 12.5 minutes. BG-aware lag selection, multi-table bounds testing and the Colombian and FX co-baselines are now in production, and the hybrid XGBoost narrows the NARDL gap by 52% without closing it (DM $p = 0.013$). Stage 1 residual serial correlation persists (BG(12) $p = 0.0003$) and is not regime-driven, and the asymmetric long-run multipliers are sensitive to break-dummy controls. Iteration 3 candidates are deferred pending Chapter 4 sign-off.

STATUS AT DRAFTING

Iteration 2: code changes implemented; pipeline run pending. Results section to be populated after run. Sections 1–5 are the pre-run record; run results follow in Section 6.

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Scope and rationale

Iteration 2 implements six methodological changes scoped during the iteration-1 review (Iteration_Reports/Iteration_1_Model_Log.docx, Section 4). The objective is to address the five known limitations of iteration 1 (BG(12) residual serial correlation at Stage 1; borderline Stage 2 cointegration verdict; FX-vs-no-FX BIC tension; under-utilisation of the structural ECT feature by Stage 1 XGBoost; Colombian Milds dominating NY on every empirical metric) while preserving the iteration-1 results as the frozen first-pass baseline. Iteration 1 is read-only under Codes/Iteration_1/. Iteration 2 lives in Codes/Iteration_2/ and is self-contained: its own Data/ snapshot, _outputs/, and code base.

The Tier 2.1 feature trim and Tiers 2.3, 2.4, 3.1, 3.2 changes from the iteration-1 plan were deferred to iteration 3 or beyond, as agreed during scope negotiation. The Tier 3.3 conditional fallback (Stage 1 NARDL with structural-break dummies as exogenous controls) will be triggered only if change 1.1 (BG-aware lag selection) fails to resolve the Stage 1 BG(12) $p < 0.05$ issue.

Change ledger

Each row below describes one change, the iteration-1 state, the iteration-2 state, and the file/function modified. Every code modification carries an inline '# ITER2 change X.X' comment so the diff between iterations is auditable line-by-line.

#	CHANGE	ITERATION 1 STATE	ITERATION 2 STATE	FILE / FUNCTION
1.1	BG-aware Stage 1 lag selection	SBC alone selected (p,q,r)=(1,0,0). BG(12) $p = 0.0001$ — strong residual serial correlation.	After SBC pick, greedily extend (p,q,r) one lag at a time (caps 12,6,4) while BG(12) $p < 0.05$. Both SBC-only and BG-augmented (p,q,r) reported.	step3a_nardl_stage1.py: fit_stage1() — BG-driven extension block; new column 'selection' in summary
1.2	Stage 2 (m=1, n=1) sensitivity + multi-table bounds	SBC selected (m,n)=(0,0). Bounds $F = 3.85$ below Narayan I(0). Inconclusive cointegration verdict.	Forced (1,1) sensitivity reported alongside SBC pick. Bounds verdict reported under Narayan Case III, Narayan Case II, and Pesaran asymptotic Case III simultaneously.	step3b_nardl_stage2.py: fit_stage2(force_lags=...); main() concatenates SBC and forced summaries
1.3	Colombian Milds as co-baseline	NY 'C' Futures was sole primary; Colombian and ICO appeared only in robustness §3.7.2.	Stage 1 NARDL run for NY+FX, NY no-FX, Col+FX, Col no-FX. All four reported as co-baselines in nardl_stage1_baselines.csv.	step3a_nardl_stage1.py: main() rewritten with 4-spec matrix
1.4	FX side-by-side reporting	FX-free was a sensitivity in §3.7.6.	Folded into 1.3 four-spec matrix; FX-included and FX-free are now co-baselines for both NY and Colombian.	step3a_nardl_stage1.py: main() (no separate code; produced as part of 1.3)
2.2	NARDL fitted-value as XGBoost feature (hybrid)	Stage 1 XGBoost feature catalogue had 12 lagged features; Stage1_ECT_lag1 mean SHAP only 0.019, suggesting underutilisation of NARDL structural information.	13th feature 'nardl_pred_t' added: the level lnAUC implied by NARDL coefficients fit on training-window data only. No leakage. Forces XGBoost to use NARDL's prediction as a starting point and learn nonlinear corrections on top.	step4a_xgb_stage1_monthly.py: feat_names, build_stage1_features() callsite; new helper _nardl_fitted_levels_for_block()
3.4	PSS Case II bounds alongside Case III	Cointegration verdict reported only under PSS Case III (unrestricted intercept, no trend).	Both Case III (unrestricted intercept) and Case II (restricted intercept) reported simultaneously, at both Stage 1 and Stage 2. Tests whether cointegration verdict is robust to the deterministic-component case as well as the critical-value table.	step3a / step3b: new PSS_CRIT tables and verdict generators

Table 1. Iteration 2 change ledger. Each change is tagged '# ITER2 change X.X' in code.

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Output files expected to differ

After the iteration-2 pipeline runs, these CSVs under Codes/Iteration_2/_outputs/tables/ should differ from their iteration-1 counterparts under Codes/Iteration_1/_outputs/tables/. A diff between the two folders forms the empirical audit trail of iteration 2.

FILE	WHAT CHANGED	HEADLINE METRIC TO COMPARE
nardl_stage1_baselines.csv (NEW)	4-spec matrix added	L^+/L^- across NY/Col × FX/no-FX
nardl_stage1_summary.csv	Adds p_sbc/q_sbc/r_sbc + selection columns	Whether (p,q,r) extended beyond (1,0,0)
nardl_stage1_diagnostics.csv	BG12_p across all four specs	Did extension drive BG12_p above 0.05?
nardl_stage1_cointegration.csv	Now 2 rows (Case III + Case II)	Does Case II flip the verdict?
nardl_stage1_coefficients_<spec>.csv (NEW)	Per-spec coefficient tables	Coefficient stability across specs
nardl_stage2_summary.csv	Now 2 rows (SBC + forced m=1,n=1)	Does richer dynamics change bounds F?
nardl_stage2_coefficients_m1n1.csv (NEW)	Forced-dynamics coefficients	—
nardl_stage2_cointegration.csv	Now 3 rows (Narayan III + II + Pesaran III)	Does any spec/table flip the verdict?
xgb_stage1_metrics.csv	Now 13-feature hybrid model	Did RMSE log drop below NARDL's 0.147?
xgb_stage1_dm.csv	DM HLN with hybrid XGBoost	Did p-value rise above 0.05?
xgb_stage1_shap_top.csv	Now ranks 13 features incl. nardl_pred_t	Does nardl_pred_t dominate the catalogue?
xgb_stage1_predictions.csv	Hybrid XGBoost predictions	Year-by-year MAE comparison

Table 2. Output tables expected to change between Codes/Iteration_1/ and Codes/Iteration_2/.

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Results review plan (pre-run)

After running 'python run_all.py' in Codes/Iteration_2/, this section will be populated with: (a) iteration-2 vs iteration-1 headline metrics, (b) change-by-change attribution for the observed shifts, (c) a consolidated answer to whether each iteration-1 known limitation has been resolved, and (d) iteration-3 candidate refinements.

Place-holders for the four headline quantities to be filled after the run:

QUANTITY	ITERATION 1 VALUE	ITERATION 2 VALUE	Δ	VERDICT
Stage 1 NARDL BG(12) p (NY+FX)	0.0001	(after run)	—	(after run)
Stage 1 selected (p,q,r)	(1,0,0)	(after run)	—	(after run)
Stage 2 bounds F under (m=1,n=1)	n/a	(after run)	—	(after run)

QUANTITY	ITERATION 1 VALUE	ITERATION 2 VALUE	Δ	VERDICT
Stage 2 cointegration verdict	inconclusive	(after run)	—	(after run)
Stage 1 NARDL→XGBoost RMSE log gap	0.0357	(after run)	—	(after run)
Stage 1 hybrid XGBoost vs NARDL DM p	< 0.0001 (XGB loses)	(after run)	—	(after run)
Colombian vs NY: BIC gap (Stage 1)	-3.8 (Col preferred)	(after run)	—	(after run)

Table 3. Pre-run placeholder tracker. Completed values are in Table 4.

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Iteration 3 candidates (pre-run)

Pending. Will be populated after iteration 2 results review identifies which iteration-1 limitations remain unresolved and which new questions iteration 2 raised. Strong a priori candidates for iteration 3 include: (a) feature-trim Stage 1 XGBoost using out-of-sample SHAP within rolling folds (deferred Tier 2.1); (b) recess-month auction interpolation (deferred Tier 3.2); (c) true Bai-Perron sequential supF in Python (deferred Tier 3.1); (d) any new limitation surfaced by iteration 2 results.

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Iteration 2 run results

The iteration-2 pipeline ran end-to-end in 12.5 minutes. All 8 steps completed. The iteration-2.1 break-dummy diagnostic was then run as a separate 5-second script. Headline results:

6.1 Headline metric tracker

QUANTITY	ITERATION 1	ITERATION 2	VERDICT
Stage 1 NARDL (NY+FX) lag spec	(p,q,r)=(1,0,0) SBC-only	(1,2,0) BG-augmented	BG-aware extension fired
Stage 1 NARDL (NY+FX) BG(12) p	0.0001	0.0003	marginal improvement; still rejecting
Stage 1 bounds F (NY+FX)	16.95	18.07	stronger cointegration
Stage 1 L+ / L- (NY+FX)	0.825 / 0.697	0.834 / 0.704	asymmetry gap unchanged at 0.13
Stage 1 Wald LR p (asymmetry)	0.0002	0.0001	asymmetry decisively supported in baseline
Stage 1 Wald SR p (asymmetry)	0.0105	0.0573	softened to borderline due to wider q span
Stage 2 SBC pick	(0,0)	(0,0)	unchanged

QUANTITY	ITERATION 1	ITERATION 2	VERDICT
Stage 2 forced (1,1) bounds F	n/a	2.69 (vs 3.85 SBC)	richer dynamics worse; SBC pick preferred
Stage 2 cointegration verdict	inconclusive (1 table)	inconclusive (3 tables)	robust to spec / table choice
Stage 1 NARDL→XGBoost RMSE log gap	0.036	0.017	52% reduction; gap remains
Stage 1 hybrid XGBoost vs NARDL DM p	< 0.0001	0.013	still rejects at 5%; not at 1%
nardl_pred_t SHAP mass share	n/a	51%	dominates; structural features collapse
Colombian vs NY: BIC gap (Stage 1)	-3.8 (Col preferred)	+0.5 (essentially tied)	iter-1 finding nuanced; both defensible

Table 4. Iteration 1 vs iteration 2 headline metrics.

6.2 Iteration 2.1 break-dummy diagnostic (Tier 3.3)

The diagnostic re-estimated the iteration-2 NY+FX baseline with five Binseg-derived step dummies as exogenous controls in the level equation. The questions: (a) is the residual BG(12) serial correlation regime-driven, and (b) are the asymmetric long-run multipliers stable to break-dummy controls? The answers:

QUANTITY	ITER-2 BASELINE	ITER-2.1 + 5 DUMMIES	Δ
n_obs	296	296	0
(p,q,r)	(1,2,0)	(1,2,0)	no change
theta1 (conditional ECM)	-0.290	-0.348	more negative
L+ (positive-shock multiplier)	0.834	0.601	-0.234
L- (negative-shock multiplier)	0.704	0.622	-0.082
L+ - L- (asymmetry gap)	+0.130	-0.021	GAP COLLAPSES TO ~0
Wald LR p (asymmetry)	0.0001	0.6583	NO LONGER REJECTING
Bounds F	18.07	23.04	+5.0 (cointegration tightens)
BDM t	-7.15	-8.02	stronger
BG(12) p	0.0003	0.0035	+0.003 — STILL REJECTING at 1%
BIC	-279.0	-268.2	+10.8 — break dummies disfavoured

Table 5. Iteration-2 NY+FX baseline vs the same specification with five Binseg step dummies.

6.3 Substantive interpretation of the diagnostic

Two findings and a third tension. (1) The BG(12) residual serial correlation is NOT regime-driven; even with five break dummies absorbing the major regime-level shifts, BG(12) p moves only from 0.0003 to 0.0035 and still rejects at 1%. The residual autocorrelation reflects something else — most plausibly volatility clustering or unmodelled nonlinear dynamics that the NARDL framework cannot capture — and is reported as a known limitation of the Stage 1 specification with HAC standard errors providing partial mitigation.

KNOWN LIMITATION

Stage 1 residual serial correlation is not regime-driven: BG(12) $p = 0.0035$ with five break dummies, still rejecting at 1%. HAC standard errors provide partial mitigation only.

(2) The cointegration verdict tightens (bounds F up to 23.04), confirming that the iteration-2 cointegration finding is robust to break-dummy controls.

(3) The third finding requires careful handling. The asymmetric long-run multipliers L^+ and L^- are sensitive to break-dummy inclusion. Three interpretations are defensible: (A) iteration-2 baseline asymmetry is the true pattern and the dummies over-control because BIC strongly disfavours them (-279 vs -268 , an 11-unit penalty); (B) the diagnostic asymmetry is the true pattern and baseline is regime-confounded; (C) both are valid, with the asymmetric pass-through being more pronounced during regime-shift periods (1999 collapse, 2008–2011 boom, 2020–2022 surge) than during calm periods, so that the dummies absorb high-asymmetry variance and the residual asymmetry is the calm-regime average.

Chapter 4 adopts Interpretation A as the headline framing on Schwarz Bayesian grounds, with Interpretation C noted as the most economically plausible explanation of the diagnostic sensitivity. The break-dummy diagnostic is reported as evidence that the asymmetric pattern is more pronounced during regime-shift periods than during calm periods — a defensible substantive nuance rather than a refutation of the iteration-2 baseline finding.

6.4 Iteration audit summary

Iteration 2 succeeded at the four methodological objectives: BG-aware lag selection is now in production (change 1.1); Stage 2 cointegration verdict is reported under three critical-value tables (changes 1.2 + 3.4); Colombian Mils is a co-baseline alongside NY (change 1.3); FX-included and FX-free are co-baselines (change 1.4); the hybrid XGBoost feature is the cleanest test of structural near-sufficiency we can construct (change 2.2).

Iteration 2 partially resolved the BG(12) limitation: the BG-aware extension reduced the test statistic but did not push p above 5%, and the iteration-2.1 diagnostic showed the residual SC is not regime-driven. The known limitation is therefore framed transparently in Chapter 4 §4.3 and Chapter 3 §3.8, with HAC standard errors providing partial mitigation. This is the cleanest available outcome short of moving to a NARDL-X or volatility-aware specification, which is flagged as iteration-3 candidate work.

Iteration 2 sharpened rather than overturned iteration-1 substantive findings.

ITERATION AUDIT SUMMARY · §6.4

The asymmetric pass-through verdict, the cointegration verdicts, and the NARDL near-sufficiency claim all hold, with appropriate refinements: the asymmetry is supported in the iteration-2 NY+FX baseline ($p = 0.0001$) but break-dummy-sensitive ($p = 0.66$ with five dummies), Stage 2 cointegration remains inconclusive across three critical-value tables, and the hybrid XGBoost narrows the NARDL gap by 52% without closing it ($DM\ p = 0.013$).

6.5 Iteration 3 — candidate refinements

Three candidate refinements emerge from iteration 2 results:

-  3.1 (deferred Tier 2.1): out-of-sample-SHAP-driven Stage 1 feature selection within rolling-origin folds. The iteration-2 hybrid SHAP shows `nardl_pred_t` carrying 51% of feature mass and structural features collapsing — the catalogue could be trimmed.
-  3.2 (new): GARCH-aware Stage 1 NARDL or NARDL-X specification to address the non-regime-driven BG(12) residual autocorrelation surfaced by iteration 2.1.
-  3.3 (deferred Tier 3.1): true Bai-Perron sequential supF in pure Python, replacing the ruptures Binseg approximation. Substantive verdicts unlikely to change but methodological credibility improves.

 END OF REPORT